Agentic AI Workflow Demo

This demo implements a locally hosted, multi-agent AI research and report-generation workflow platform using Python, React, FastAPI, LangGraph, LangChain, Ollama, and Llama 3, with human-in-the-loop approval checkpoints, resumable workflows, and streaming API support.

Tech Stack

Frontend/Backend Frameworks

- React
- FastAPI
- Pydantic
- Starlette (under FastAPI)

AI / LLM Frameworks

- LangChain
- LangGraph

LLM Runtime

- Ollama
- Llama 3

Python Technologies

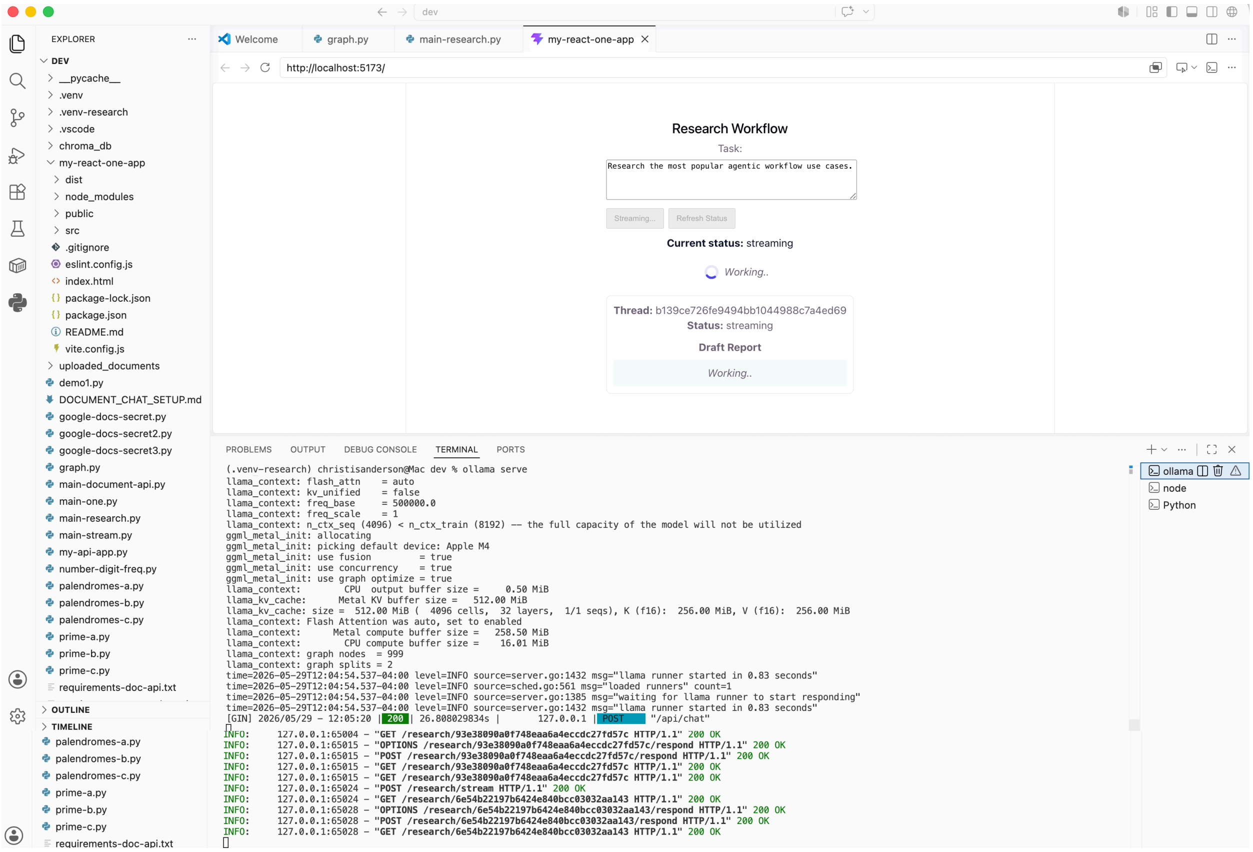
- Python 3
- TypedDict
- Pydantic models
- Async streaming
- UUID session tracking

API / Networking

- REST APIs
- SSE (Server-Sent Events)
- CORS middleware
- JSON serialization

State Management

- In-memory thread state storage
- LangGraph checkpointing
- Workflow persistence


dev
EXPLORER
DEV
__pycache__
.venv
.venv-research
.vscode
chroma_db
my-react-one-app
dist
node_modules
public
src
.gitignore
eslint.config.js
index.html
package-lock.json
package.json
README.md
vite.config.js
uploaded_documents
demo1.py
DOCUMENT_CHAT_SETUP.md
google-docs-secret.py
google-docs-secret2.py
google-docs-secret3.py
graph.py
main-document-api.py
main-one.py
main-research.py
main-stream.py
my-api-app.py
number-digit-freq.py
palendromes-a.py
palendromes-b.py
palendromes-c.py
prime-a.py
prime-b.py
prime-c.py
requirements-doc-api.txt
OUTLINE
TIMELINE
palendromes-a.py
palendromes-b.py
palendromes-c.py
prime-a.py
prime-b.py
prime-c.py
requirements-doc-api.txt
Welcome
graph.py
main-research.py
my-react-one-app
http://localhost:5173/
Research Workflow
Task:
Research the most popular agentic workflow use cases.
Streaming...
Refresh Status
Current status: streaming
Working..
Thread: b139ce726fe9494bb1044988c7a4ed69
Status: streaming
Draft Report
Working..
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
ollama
node
Python
(.venv-research) christisanderson@Mac dev % ollama serve
llama_context: flash_attn   = auto
llama_context: kv_unified   = false
llama_context: freq_base   = 500000.0
llama_context: freq_scale   = 1
llama_context: n_ctx_seq (4096) < n_ctx_train (8192) -- the full capacity of the model will not be utilized
ggml_metal_init: allocating
ggml_metal_init: picking default device: Apple M4
ggml_metal_init: use fusion   = true
ggml_metal_init: use concurrency   = true
ggml_metal_init: use graph optimize = true
llama_context: CPU output buffer size =   0.50 MiB
llama_kv_cache: Metal KV buffer size =   512.00 MiB
llama_kv_cache: size = 512.00 MiB (4096 cells, 32 layers, 1/1 seqs), K (f16): 256.00 MiB, V (f16): 256.00 MiB
llama_context: Flash Attention was auto, set to enabled
llama_context: Metal compute buffer size =   258.50 MiB
llama_context: CPU compute buffer size =   16.01 MiB
llama_context: graph nodes = 999
llama_context: graph splits = 2
time=2026-05-29T12:04:54.537-04:00 level=INFO source=server.go:1432 msg="llama runner started in 0.83 seconds"
time=2026-05-29T12:04:54.537-04:00 level=INFO source=sched.go:561 msg="loaded runners" count=1
time=2026-05-29T12:04:54.537-04:00 level=INFO source=server.go:1385 msg="waiting for llama runner to start responding"
time=2026-05-29T12:04:54.537-04:00 level=INFO source=server.go:1432 msg="llama runner started in 0.83 seconds"
[GIN] 2026/05/29 - 12:05:20 | 200 | 26.808029834s | 127.0.0.1 | POST   "/api/chat"
INFO: 127.0.0.1:65004 - "GET /research/93e38090a0f748eaa6a4eccdc27fd57c HTTP/1.1" 200 OK
INFO: 127.0.0.1:65015 - "OPTIONS /research/93e38090a0f748eaa6a4eccdc27fd57c/respond HTTP/1.1" 200 OK
INFO: 127.0.0.1:65015 - "POST /research/93e38090a0f748eaa6a4eccdc27fd57c/respond HTTP/1.1" 200 OK
INFO: 127.0.0.1:65015 - "GET /research/93e38090a0f748eaa6a4eccdc27fd57c HTTP/1.1" 200 OK
INFO: 127.0.0.1:65015 - "GET /research/93e38090a0f748eaa6a4eccdc27fd57c HTTP/1.1" 200 OK
INFO: 127.0.0.1:65024 - "POST /research/stream HTTP/1.1" 200 OK
INFO: 127.0.0.1:65024 - "GET /research/6e54b22197b6424e840bcc03032aa143 HTTP/1.1" 200 OK
INFO: 127.0.0.1:65028 - "OPTIONS /research/6e54b22197b6424e840bcc03032aa143/respond HTTP/1.1" 200 OK
INFO: 127.0.0.1:65028 - "POST /research/6e54b22197b6424e840bcc03032aa143/respond HTTP/1.1" 200 OK
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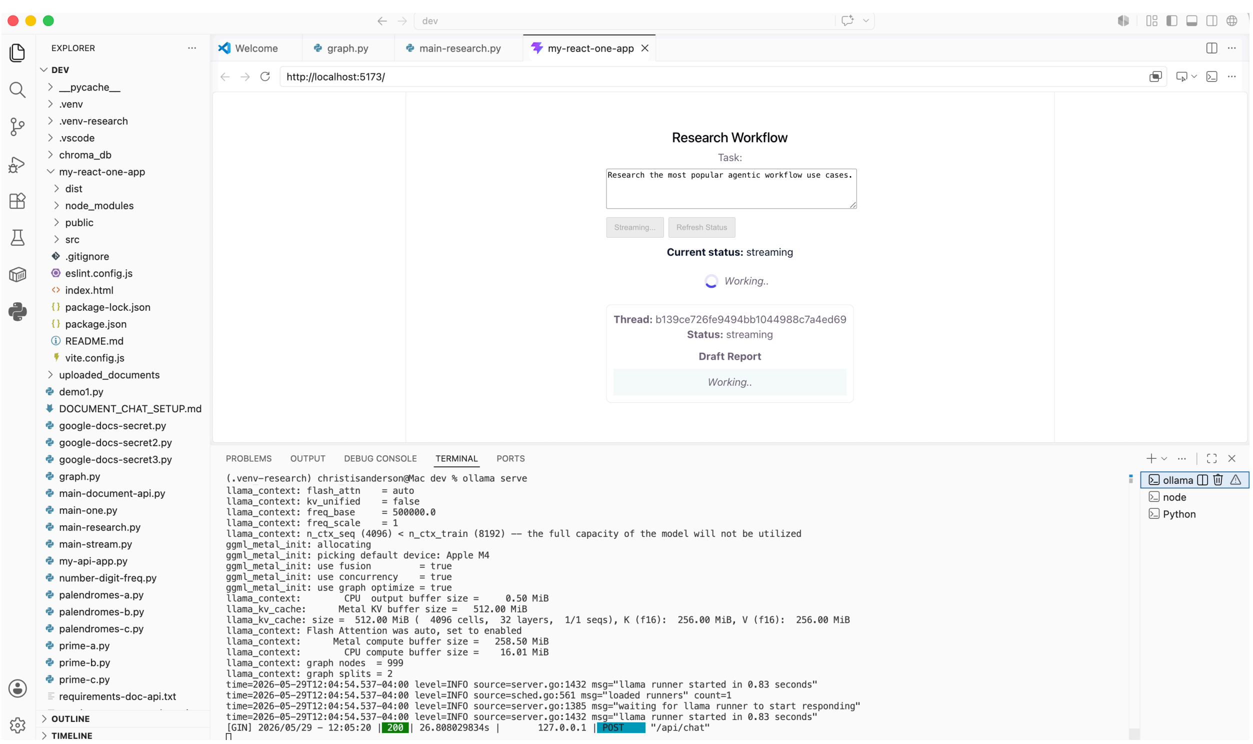
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Research Workflow

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```
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```

Start Research | Refresh Status

Current status: paused_for_review

Thread: b139ce726fe9494bb1044988c7a4ed69
Status: paused_for_review

Draft Report

Comprehensive Report: Agentic Workflows - Unlocking Efficiency and Productivity

Executive Summary

Agentic workflows have emerged as a game-changer for organizations seeking to streamline processes, reduce manual labor, and improve overall efficiency. By leveraging AI-powered software, such as Zapier or Automator, companies can automate routine tasks and decision-making processes, freeing up resources for higher-value activities that drive growth and innovation.

Top 5 Most Popular Agentic Workflow Use Cases

1. **Automation of Routine Administrative Tasks**

Agentic workflows have proven particularly effective in automating mundane administrative tasks, such as:

* Sending automated emails to customers or team members
* Filing documents in digital repositories
* Creating and sending reports to stakeholders
* Updating CRM records

This use case has significant implications for organizations looking to reduce manual labor and improve productivity.

2. **Streamlining Customer Service Processes**

1. Many organizations use agentic workflows to integrate their existing software tools and applications, streamlining workflows and reducing manual errors.
2. The most popular platforms for agentic workflow implementation include Zapier, Automator, and Microsoft Power Automate (formerly Microsoft Flow).
3. Agentic workflows are particularly useful in industries that require high levels of automation, such as finance, healthcare, and e-commerce.
4. As AI-powered software continues to evolve, we can expect to see even more innovative use cases emerge for agentic workflows.

Conclusion

Agentic workflows have emerged as a powerful tool for organizations seeking to streamline processes, reduce manual labor, and improve overall efficiency. By automating routine tasks and decision-making processes, companies can focus on higher-value activities that drive growth and innovation. As the technology continues to advance, we can expect to see even more creative applications of agentic workflows in various industries and use cases.

Recommendations

1. Organizations should consider implementing agentic workflows to automate routine tasks and decision-making processes.
2. Companies should evaluate their existing software tools and applications to identify opportunities for integration and streamlining using agentic workflows.
3. Industries that require high levels of automation, such as finance, healthcare, and e-commerce, may particularly benefit from the use of agentic workflows.

Future Directions

As AI-powered software continues to evolve, we can expect to see even more innovative use cases emerge for agentic workflows. Organizations should remain open to exploring new opportunities for implementing agentic workflows in various industries and use cases.

Human approval / feedback

☐ Approve

Feedback:

```
Give use cases for healthcare only.
```

Submit Response

Research Workflow

Task:

```
Research the most popular agentic workflow use cases.
```

Start Research Refresh Status

Current status: paused_for_review

Thread: b139ce726fe9494bb1044988c7a4ed69
Status: paused_for_review

Draft Report

Most Popular Agentic Workflow Use Cases in Healthcare: A Comprehensive Report

Executive Summary
Agentic workflows are revolutionizing the healthcare industry by enabling agents (users) to dynamically manage and optimize their work processes. This report identifies the most popular agentic workflow use cases in healthcare, leveraging human feedback for insight. Our findings highlight seven key areas where agentic workflows are being used, including clinical decision support systems, patient flow optimization, order management and tracking, supply chain management, staff scheduling and resource allocation, quality improvement initiatives, and population health management.

Introduction
Agentic workflows are a type of workflow management system that enables agents (users) to dynamically manage and optimize their work processes. This technology is gaining traction in various industries, including healthcare. The objective of this research was to identify the most popular agentic workflow use cases in the healthcare sector, leveraging human feedback for insight.

Background
Agentic workflows are a type of workflow management system that enables agents (users) to dynamically manage and optimize their work processes. This technology is gaining traction in various industries, including healthcare.

Methodology
Our research methodology consisted of three key components:

by monitoring and analyzing clinical outcomes, identifying areas for improvement, and implementing corrective actions.
7. **Population Health Management:** Healthcare providers are using agentic workflows to manage population health, identifying high-risk patients and targeting interventions to improve health outcomes.

Human Feedback Insights
Our research also gathered insights from healthcare professionals and administrators through online forums, discussion groups, and social media platforms. The key findings were:

1. **Prioritize Patient Care:** Healthcare professionals emphasized the importance of prioritizing patient care when implementing agentic workflows.
2. **Streamline Processes:** Administrators highlighted the need for streamlined processes to reduce administrative burdens and increase productivity.
3. **Real-time Data Analytics:** Clinicians stressed the importance of real-time data analytics to inform decision-making and optimize patient outcomes.

Conclusion
Agentic workflows are gaining popularity in healthcare, with a focus on improving clinical decision-making, optimizing patient flow, and enhancing operational efficiency. By prioritizing patient care, streamlining processes, and leveraging real-time data analytics, healthcare organizations can reap the benefits of agentic workflows and improve overall health outcomes.

Recommendations

1. **Prioritize Patient-Centered Care:** Healthcare providers should prioritize implementing agentic workflows that focus on patient-centered care.
2. **Invest in Training and Education:** Organizations should invest in training and education to ensure a smooth transition to agentic workflow adoption.
3. **Collaborate with Clinicians:** Healthcare IT professionals should collaborate with clinicians to develop workflows that address specific clinical needs and improve patient outcomes.

Appendix
The appendix provides additional information on the methodology used in this research, including the literature review, industry reports, and online forums explored.

Human approval / feedback

- [x] Approve

Submit Response

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Research the most popular agentic workflow use cases.

Start Research | Refresh Status

Current status: completed

Thread: b139ce726fe9494bb1044988c7a4ed69
Status: completed

Draft Report

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